

Agent Based on Propositional Logic

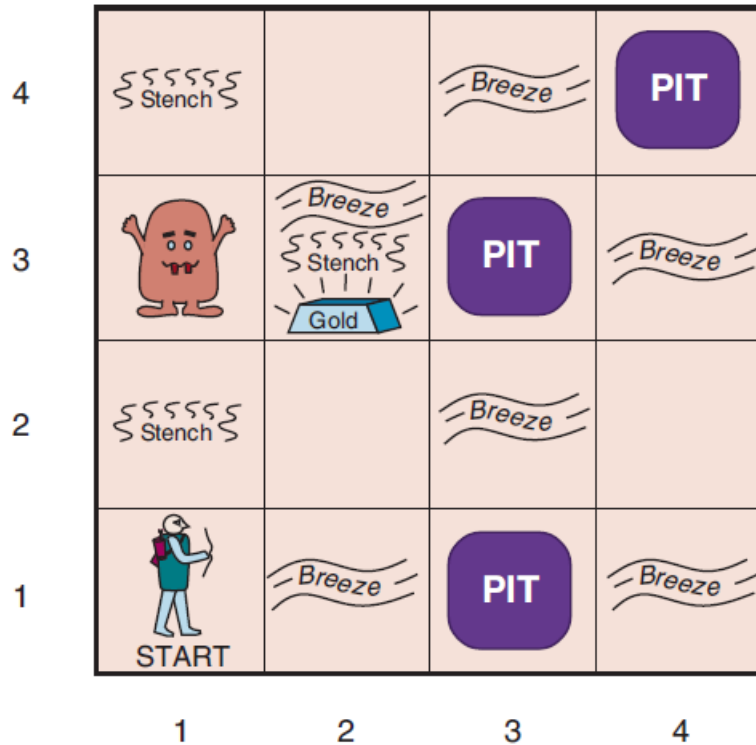
- Write down a complete **logical model** of the effects of action.
- How **logical inference** can be used by an agent?.
- How to keep track of the world without resorting to **inference** history?
- How to use **logical inference** to construct plans based on the KB?

Knowledge base (KB):

♣ general knowledge about how the world works

♣ percept sentences obtained in a particular world

Current State in the Wumpus World



Axioms (given without being derived):

$$\neg P_{1,1}$$

$$\neg W_{1,1}$$

$$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$$

// 16 rules of this type

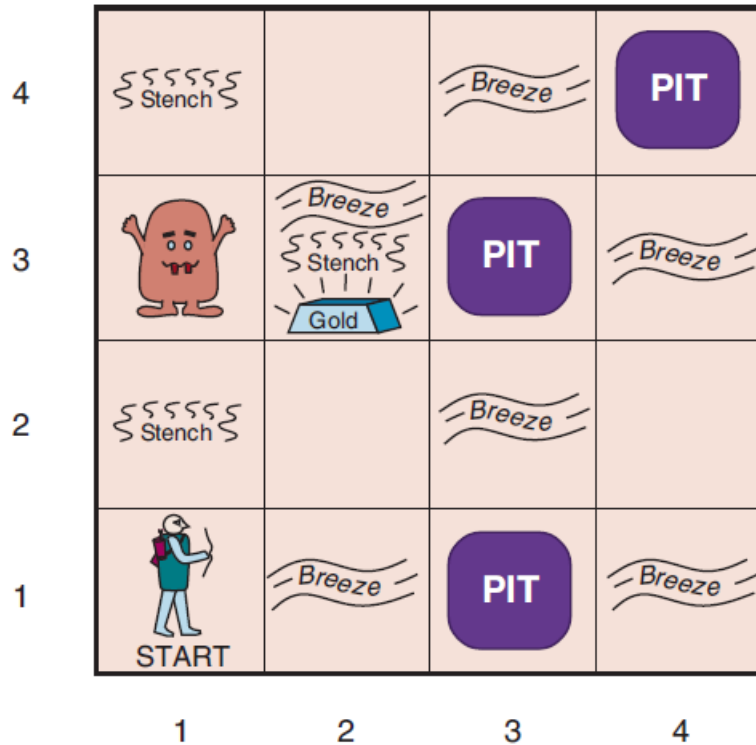
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- $P_{x,y} = \text{true}$ if there is a pit in $[x, y]$.
- $W_{x,y} = \text{true}$ if there is a wumpus in $[x, y]$, dead or alive.
- $B_{x,y} = \text{true}$ if the agent perceives a breeze in $[x, y]$.
- $S_{x,y} = \text{true}$ if the agent perceives a stench in $[x, y]$.

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♦ Exactly one wumpus

$$W_{1,1} \vee W_{1,2} \vee \dots \vee W_{4,3} \vee W_{4,4}$$

// ≥ 1 wumpus

$$\neg W_{i,j} \vee \neg W_{k,l}$$

$$1 \leq i, j, k, l \leq 4 \text{ and } (i, j) \neq (k, l)$$

// ≤ 1 Wumpus;

$$// \binom{16}{2} = \frac{16 \times 15}{2} = 120 \text{ rules}$$

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Representing Percepts

A percept asserts something only about the current time.

- $Stench^4$: the agent senses stench at time step 4 (in square A).
- $\neg Stench^3$: the agent senses no stench at time step 3 (in square B).

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Associate propositions with time steps for aspects of the world that changes over time.

$L_{1,1}^0$: the agent is in square $[1, 1]$ at time step 0.

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$$L_{x,y}^t \Rightarrow (\textit{Breeze}^t \Leftrightarrow B_{x,y})$$

$$L_{x,y}^t \Rightarrow (\textit{Stench}^t \Leftrightarrow S_{x,y})$$

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fluent — $L_{x,y}^t \Rightarrow (Breeze^t \Leftrightarrow B_{x,y})$
(aspect changing with time) $L_{x,y}^t \Rightarrow (Stench^t \Leftrightarrow S_{x,y})$

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Effect axioms specify outcome of an action at the next time step.

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// if the agent is at [1,1] facing east at time 0 and goes forward,
// the result is that the agent is in [2,1] and no longer in [1,1].

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$$Forward^t \Rightarrow (HaveArrow^t \Leftrightarrow HaveArrow^{t+1})$$

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$O(mn)$ frame axioms for m actions and n fluents

Axioms for Successor States

Successor-state axiom, one for every fluent F , states that

- either the action at t causes F to be true at $t + 1$,
- or F was already true at t and the action does not cause it to be false.

$$F^{t+1} \Leftrightarrow \text{ActionCauses}F^t \vee (F^t \wedge \neg \text{ActionCausesNot}F^t)$$

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The diagram shows three vertical lines connecting the terms in the general axiom to the specific axiom below. The first line connects F^{t+1} to HaveArrow^{t+1} . The second line connects $\text{ActionCauses}F^t$ to HaveArrow^t . The third line connects $\neg \text{ActionCausesNot}F^t$ to $\neg \text{Shoot}^t$.

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$$L_{1,1}^{t+1} \Leftrightarrow (L_{1,1}^t \wedge (\neg \text{Forward}^t \vee \text{Bump}^{t+1})) \vee (L_{1,2}^t \wedge (\text{FacingSouth}^t \vee \text{Forward}^t)) \\ \vee (L_{2,1}^t \wedge (\text{FacingWest}^t \vee \text{Forward}^t))$$

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Square-OK axiom asserts that a square is free of a pit or live Wumpus.

$$OK_{x,y}^t \Leftrightarrow \neg P_{x,y} \wedge \neg (W_{x,y} \wedge \text{WumpusAlive}^t)$$

Initial Percepts and Actions

$\neg \text{Stench}^0 \wedge \neg \text{Breeze}^0 \wedge \neg \text{Glitter}^0 \wedge \neg \text{Bump}^0 \wedge \neg \text{Scream}^0$; FacingEast⁰; Forward⁰
 $\neg \text{Stench}^1 \wedge \text{Breeze}^1 \wedge \neg \text{Glitter}^1 \wedge \neg \text{Bump}^1 \wedge \neg \text{Scream}^1$; TurnRight¹
 $\neg \text{Stench}^2 \wedge \text{Breeze}^2 \wedge \neg \text{Glitter}^2 \wedge \neg \text{Bump}^2 \wedge \neg \text{Scream}^2$; TurnRight²
 $\neg \text{Stench}^3 \wedge \text{Breeze}^3 \wedge \neg \text{Glitter}^3 \wedge \neg \text{Bump}^3 \wedge \neg \text{Scream}^3$; Forward³
 $\neg \text{Stench}^4 \wedge \neg \text{Breeze}^4 \wedge \neg \text{Glitter}^4 \wedge \neg \text{Bump}^4 \wedge \neg \text{Scream}^4$; TurnRight⁴
 $\neg \text{Stench}^5 \wedge \neg \text{Breeze}^5 \wedge \neg \text{Glitter}^5 \wedge \neg \text{Bump}^5 \wedge \neg \text{Scream}^5$; Forward⁵
 $\text{Stench}^6 \wedge \neg \text{Breeze}^6 \wedge \neg \text{Glitter}^6 \wedge \neg \text{Bump}^6 \wedge \neg \text{Scream}^6$

Query the knowledge base:

$\text{Ask}(KB, L_{1,2}^6) = \text{true}$

$\text{Ask}(KB, W_{1,3}) = \text{true}$

$\text{Ask}(KB, P_{3,1}) = \text{true}$

$\text{Ask}(KB, OK_{2,2}^6) = \text{true}$

// the square [2,2] is OK to move into.

1,4	2,4 P?	3,4	4,4
1,3 W!	2,3 A S G B	3,3 P?	4,3
1,2 S V OK	2,2 V OK	3,2	4,2
1,1 V OK	2,1 B V OK	3,1 P!	4,1